

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B.Tech Examination (CE/IT)

May 2014

IT206.01 Object Oriented Programming using JAVA

Date: 03.05.2014, Saturday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks:70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

- (a) What is Byte code? How the execution of java program is different from any other programming language? 3
- (b) Create a class called vehicle with two attribute break and horn. Create three sub classes called BUS, CYCLE, and TRUCK with attribute seats, spokes, luggage_pressure respectively. Display all the attributes of all class using super keyword. 4

Q-2

- (a) Java is a strongly typed language. Justify. 2
- (b) Explain the meaning and working of *garbage collection*, *this* keyword and *run time polymorphism*. 7
- (c) Compare final, finally and finalize () with example. 5

OR

Q-2

- (a) State whether following statements are True or False with Justification. 5
 1. Interface can be instantiated.
 2. A package statement can occur either before or after import statement.
 3. Final class cannot be extended.
 4. A constructor cannot have private scope.
 5. Declaration of variable is possible in interface.
- (b) Demonstrate the use of command line argument program with example and also explain the difference between sleep () and wait () of multithreading. 4
- (c) Write a program to create a class called box with three attributes width, height and depth. Calculate the volume of box using constructor overloading concept. 5

Q-3 Attempt Any Two.

- (a) What is casting? Explain type conversion and type casting in detail with example. Also explain any two bitwise operators in detail. 14
- (b) Write a program to convert given binary number into decimal number.
- (c) Write a program to implement the concept of abstract class and abstract methods.

SECTION-II

Q-4

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|-----|--|---|
| (a) | What is adapter class? Explain with example. | 4 |
| (b) | Differentiate between Object-Oriented (OO) development approach and structured development approach. | 3 |

Q-5

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|-----|---|---|
| (a) | Write a program to create an applet with three buttons named as "Round Rectangle", "Arc", and "Circle". By clicking on that button particular shape will be drawn in applet window with color "blue". | 3 |
| (b) | Identify at least six classes of university library and create a class hierarchy to organize classes. | 6 |
| (c) | What is operation overriding? Explain how overriding is used for following : | 5 |
| | 1. Extension | |
| | 2. Restriction | |
| | 3. Optimization | |
| | 4. Convenience | |

OR

Q-5

- | | | |
|-----|---|---|
| (a) | Fill in the Blanks. | 3 |
| | 1. _____ diagram is good at describing the behavior of an object across several use cases. | |
| | 2. _____ diagram is a valuable tool to help understand the functional requirements of a system. | |
| | 3. _____ diagram should be used when you want to look at the behavior of several objects within a single use case. | |
| (b) | In an organization, projects are divided into activities, and the activities into tasks. A task is assigned to a worker, who could be a salaried worker or an hourly worker. Each task requires a certain skill, and workers have various skills at various level of expertise. Using Noun Phrase Approach, identify list of candidate classes, relationship and draw object diagram. | 6 |
| (c) | Can states react to events without transition in state chart diagram? If Yes give example, if No, explain Why? | 5 |

Q-6 Attempt Any Two.

14

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|-----|---|--|
| (a) | What is Activity Diagram? Explain the elements of activity diagram with significance. How activity diagram is different from state chart diagram? | |
| (b) | Draw the state chart diagram for Bank ATM Machine by showing at-least 7 states of an ATM Machine. | |
| (c) | In online ticket reservation system, the passenger has to select the origin, date of journey, passport number, etc. The reservation counter keeps track of the passenger's information. The system will have all the details about the airlines and the facilities provided by them. There are various airlines provided according to the convenience of the passenger. A database is maintains by the database administrator. The passenger can select any airlines from many varieties of airlines according to their convenience. The journey can be within the country or across the country. For international passengers, the tickets are booked only after checking their visa and passport. The vacancy of each airline can be viewed online.
Identify Actors, Use Cases and Draw the Use Case Diagram for Online Ticket Reservation System. | |